Universal Coset Field Theory: An E_6-Albert-Algebra Framework

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Abstract

We formulate Universal Coset Field Theory (UCFT) as a mathematical-physics framework whose algebraic starting point is the complexified Albert algebra. The paper follows the structural arc of the original UCFT proposal but separates theorem, postulate, and conjecture. The algebraic core is a conditional spontaneous-symmetry-breaking result: for a stated class of compact E_6-invariant potentials on $J_3(\mathbb{O})_{\mathbb{C}}$, the projective vacuum is the closed rank-one orbit

$$\mathrm{EIII} = E_6/(\mathrm{Spin}(10) \cdot \mathrm{U}(1)).$$

The 32 real tangent directions are clock fields carrying $\mathfrak{m}_{\mathbb{C}} \cong \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$, and the coset supplies a composite $\mathrm{Spin}(10) \cdot \mathrm{U}(1)$ connection. The route to four-dimensional physics then requires explicit model data: a soldering field choosing a Lorentzian $H_2(\mathbb{C})$ block, a chiral Spin(10) matter sector, a standard GUT breaking chain down to the Standard Model, and an effective action whose massive modes induce an Einstein-Hilbert term in the Sakharov sense. Thus UCFT is presented as a conditional Albert-algebra embedding of Standard-Model-plus-gravitation data, not as a from-nothing derivation of those data.

Keywords. Albert algebra; complex Cayley plane; E_6; clock fields; Spin(10) grand unification; induced gravity; Standard Model embedding.

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1 Introduction

Exceptional structures are a natural arena for unification. The exceptional Jordan algebra $J_3(\mathbb{O})$ has dimension 27, its automorphism group is the compact F_4, and its complexification carries the minimal complex representation of the complex Lie group $E_6^{\mathbb{C}}$. Under

$$H = \mathrm{Spin}(10) \cdot \mathrm{U}(1)$$

the fundamental and adjoint representations branch as

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}, \tag{1}$$

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}. \tag{2}$$

The appearance of a Spin(10) spinor is the central representation-theoretic motivation for UCFT.

The purpose of this paper is more modest than earlier drafts. We do not claim that the Standard Model, Lorentzian signature, or quantum gravity follow from two axioms. We instead state:

- an algebraic setup based on $J_3(\mathbb{O})_{\mathbb{C}}$ and the compact real form of E_6;
- a conditional theorem selecting the projective rank-one orbit EIII;
- a coset sigma-model geometry with a positive-definite target metric and 32 clock-field directions;
- additional postulates that solder a four-dimensional Lorentzian spacetime and add a Spin(10) matter/Higgs sector descending to the Standard Model;
- conjectural or effective-field-theory statements about induced gravity, renormalization-group flow, and heterotic/string correspondences.

1.1 Status of claims

We use the following labels throughout.

Theorem. A statement proved from stated mathematical hypotheses.

Assumption/Postulate. Additional structure defining the model.

Conjecture. A motivated physical claim not established by the present arguments.

Estimate. A one-loop or effective-field-theory calculation whose value depends on regulator, truncation, or field-content choices.

The main logical ledger is:

Claim	Content	Status
Real-form separation	compact E_6, $E_{6(-26)}$, and $E_{6(-14)}$ are distinct	theorem/standard
Vacuum selection	stated potential selects projective rank-one orbit EIII	theorem, conditional on V
Coset tangent	$\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$	theorem/standard
Coset metric	unique H-invariant positive-definite metric	theorem
Lorentzian $d = 4$	soldered from a chosen $H_2(\mathbb{C})$ block	postulate
Matter families	one chiral 16, three generations, Yukawa sector	postulate
Induced gravity	Sakharov coefficient from massive fields	estimate
Asymptotic safety	gravitational fixed point in a truncation	conjecture
Vortex/heterotic sector	worldsheet and moduli correspondence	conjecture

2 Axioms and Model Data

The old UCFT draft began with two axioms and then asserted that the Standard Model and gravity followed. The revised formulation keeps the same motivating direction but separates axioms from model-building input.

Assumption 2.1 (Albert-algebra arena)**.** The primary order parameter is a projective field with values in the projectivized complex Albert algebra,

$$[\Phi] \in \mathbb{P}(J_3(\mathbb{O})_{\mathbb{C}}).$$

The compact real form of E_6 acts on this representation, and the vacuum is selected dynamically by an E_6-invariant potential.

Assumption 2.2 (Effective quantum dynamics)**.** Below a cutoff Λ, the theory is described by a local effective action for $[\Phi]$, a soldering field, gauge fields, and the matter multiplets specified below. Quantization is understood in the ordinary effective-field theory sense. Reflection positivity, UV completion, and nonperturbative existence are not assumed to be proved by the algebraic construction.

Postulate 2.3 (Model data beyond the algebraic arena)**.** UCFT contains the following additional structure.

P1. An E_6-invariant potential selecting the projective rank-one orbit EIII.

P2. A soldering field selecting a four-dimensional Lorentzian subspace $H_2(\mathbb{C}) \subset H_2(\mathbb{O})$.

P3. One chiral **16** of Spin(10) per family.

P4. A family sector giving three generations.

P5. A conventional Spin(10) Higgs/Yukawa sector that breaks to the Standard Model, generates fermion masses, and makes the extra abelian gauge factors heavy or otherwise absent from the low-energy spectrum.

Theorem 2.4 (Conditional UCFT embedding)**.** *Assume Theorem 2.1, Theorem 2.2, and Theorem 2.3. Then the resulting low-energy framework contains:*

(i) *a vacuum coset*

$$\mathrm{EIII} = E_6/(\mathrm{Spin}(10)\cdot \mathrm{U}(1)),$$

with 32 *real clock-field directions;*

(ii) *a composite* $\mathrm{Spin}(10)\cdot\mathrm{U}(1)$ *connection from the Maurer-Cartan form;*

(iii) *a* Spin(10) *grand-unified matter embedding containing the Standard Model representation content after the usual GUT descent;*

(iv) *an effective Lorentzian four-dimensional field theory after soldering;*

(v) *a Sakharov-type induced Einstein-Hilbert term once massive fields are integrated out.*

Proof. Item (i) is Theorem 4.3. Item (ii) is the standard connection part of the Cartan decomposition on E_6/H, discussed in subsection 6.1. Item (iii) follows from the branching (1) together with the matter and Higgs postulates in subsection 6.2 and subsection 6.4. Item (iv) is precisely the soldering postulate of Theorem 5.1. Item (v) is the effective one-loop estimate in subsection 7.3. The theorem is therefore a conditional embedding theorem, not a proof that the Standard Model and gravity are forced by the Albert algebra alone. □

3 Algebraic Background

3.1 The Albert algebra

Let $\mathbb{O}$ be the real division algebra of octonions. The Albert algebra is

$$J_3(\mathbb{O}) = \left\{ \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix} : \xi_i \in \mathbb{R},\ x_i \in \mathbb{O} \right\},$$

with Jordan product $X \circ Y = \frac{1}{2}(XY + YX)$. It is a 27-dimensional formally real Jordan algebra. The trace form and cubic norm are

$$\langle X, Y\rangle = \mathrm{Tr}(X \circ Y), \qquad Q(X) = \langle X, X\rangle, \qquad N(X) = \det X.$$

There is a quadratic adjoint, or sharp map, $X \mapsto X^\#$, characterized by

$$X^\# \circ X = N(X)I, \qquad (X^\#)^\# = N(X)X.$$

In a Jordan frame,

$$\mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)^\# = \mathrm{diag}(\lambda_2\lambda_3, \lambda_3\lambda_1, \lambda_1\lambda_2).$$

Lemma 3.1 (Rank-one criterion)**.** *For* $X \in J_3(\mathbb{O})$*,* $X^\# = 0$ *if and only if* $\mathrm{rank}\, X \le 1$.

Proof. Every element of the formally real Albert algebra is diagonalizable by the F_4-action. In a Jordan frame, $X^\# = 0$ means $\lambda_i\lambda_j = 0$ for every $i \ne j$, so at most one eigenvalue is nonzero. This is exactly $\mathrm{rank}\, X \le 1$. □

3.2 Real forms

The following distinction is essential.

- $\mathrm{Aut}(J_3(\mathbb{O})) = F_{4(-52)}$, the compact real form of F_4. It preserves the Jordan product, Q, and N.
- The reduced structure group of the real Albert algebra is $E_{6(-26)}$. It preserves N up to a real character. It does not preserve the Euclidean trace form Q.
- The compact real form $E_{6(-78)}$ acts unitarily on the complex representation $J_3(\mathbb{O})_{\mathbb{C}}$. This is the real form used for the compact projective orbit EIII.
- The noncompact Hermitian real form $E_{6(-14)}$ has maximal compact subgroup $\mathrm{Spin}(10) \cdot \mathrm{U}(1)$. It is not equivalent to compact E_6. Its symmetric space is the noncompact dual of EIII.

Remark 3.2. Earlier drafts conflated compact E_6 with $E_{6(-14)}$. This is false. In this manuscript, "E_6" without a real-form subscript means the compact real form when discussing the vacuum orbit EIII.

3.3 The complex Cayley plane

Let $J_3(\mathbb{O})_{\mathbb{C}} = J_3(\mathbb{O}) \otimes_{\mathbb{R}} \mathbb{C}$, and let $\mathbb{P}(J_3(\mathbb{O})_{\mathbb{C}})$ denote its projective space. A line $[\Phi] \in \mathbb{P}(J_3(\mathbb{O})_{\mathbb{C}})$ is rank one if $\Phi^{\#} = 0$ and $\Phi \neq 0$.

Theorem 3.3 (Closed rank-one orbit)**.** *The compact group E_6 acts transitively on the set of projective rank-one lines in $\mathbb{P}(J_3(\mathbb{O})_{\mathbb{C}})$. The stabilizer of a highest-weight line is* $\mathrm{Spin}(10) \cdot \mathrm{U}(1)$. *Hence the projective rank-one orbit is*

$$\mathcal{M} = E_6/(\mathrm{Spin}(10) \cdot \mathrm{U}(1)) = \mathrm{EIII}, \qquad \dim_{\mathbb{R}} \mathcal{M} = 32.$$

Proof. This is the standard highest-weight orbit of the minuscule 27-dimensional representation of E_6. The stabilizer of the highest-weight line is the maximal parabolic whose compact Levi factor is $\mathrm{Spin}(10) \cdot \mathrm{U}(1)$. Therefore the compact orbit is E_6/H. Since $\dim_{\mathbb{R}} E_6 = 78$ and $\dim_{\mathbb{R}}(\mathrm{Spin}(10) \cdot \mathrm{U}(1)) = 45 + 1 = 46$, the real dimension is 32. □

4 Spontaneous Symmetry Breaking and Clock Fields

4.1 Vacuum selection

Assumption 4.1 (Order parameter)**.** The algebraic order parameter is a projective field

$$[\Phi] : X \longrightarrow \mathbb{P}(J_3(\mathbb{O})_{\mathbb{C}}),$$

where X is the spacetime manifold after the soldering postulate of section 5. Equivalently, before projectivization one may use a nonzero lift $\Phi \in J_3(\mathbb{O})_{\mathbb{C}}$, with the overall complex phase regarded as gauge.

Let h be the positive Hermitian form on $J_3(\mathbb{O})_{\mathbb{C}}$ preserved by compact E_6. The sharp map complexifies to $J_3(\mathbb{O})_{\mathbb{C}} \to J_3(\mathbb{O})_{\mathbb{C}}$.

Assumption 4.2 (Potential class)**.** At the algebraic level, take a compact-E_6-invariant potential for a lift $\Phi \in J_3(\mathbb{O})_{\mathbb{C}}$ of the form

$$V(\Phi) = \alpha\, h(\Phi^{\#}, \Phi^{\#}) + \lambda\big(h(\Phi, \Phi) - v^2\big)^2, \qquad \alpha, \lambda, v > 0. \tag{3}$$

The physical vacuum is the projectivization of the minimum locus.

Theorem 4.3 (Conditional vacuum selection)**.** *Under Theorem 4.2, the projective vacuum is the rank-one orbit*

$$\mathcal{M} = \mathrm{EIII} = E_6/(\mathrm{Spin}(10) \cdot \mathrm{U}(1)).$$

Proof. Both summands of V are nonnegative. A global minimum satisfies

$$h(\Phi, \Phi) = v^2, \qquad h(\Phi^{\#}, \Phi^{\#}) = 0.$$

Since h is positive definite, $h(\Phi^{\#}, \Phi^{\#}) = 0$ implies $\Phi^{\#} = 0$. By the complexified rank-one criterion, Φ lies on the rank-one cone, and $h(\Phi, \Phi) = v^2$ excludes the origin. Projectivizing removes the lift and gives exactly the projective rank-one orbit of Theorem 3.3. □

Remark 4.4. The result is conditional on the potential. It does not prove that this potential is forced by the Albert algebra, nor does it prove Lorentzian spacetime or Standard Model matter.

4.2 Coset geometry

Let $H = \mathrm{Spin}(10) \cdot \mathrm{U}(1)$. The Lie algebra decomposition is

$$\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}, \qquad \mathfrak{h} = \mathfrak{spin}(10) \oplus \mathfrak{u}(1).$$

From (2),

$$\mathfrak{m}_{\mathbb{C}} \cong \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}. \tag{4}$$

As a real representation, $\mathfrak{m}$ has dimension 32 and carries the complex structure of the Hermitian symmetric space EIII.

Proposition 4.5 (Invariant metric)**.** *The compact symmetric space* EIII *has a unique E_6-invariant Riemannian metric up to scale. Its restriction to $\mathfrak{m}$ is positive definite.*

Proof. For the compact real form, the negative of the Killing form is positive definite. On the irreducible Hermitian symmetric space E_6/H, the isotropy representation is irreducible as a real representation of complex type, so Schur's lemma gives uniqueness up to scale. □

Corollary 4.6. *Lorentzian signature does not follow from the coset metric. The coset metric has signature* $(32, 0)$*.*

Definition 4.7 (Clock fields)**.** After vacuum selection, a UCFT clock-field configuration is a map

$$\phi : X \longrightarrow \mathcal{M} = \mathrm{EIII}.$$

In a local coordinate chart on $\mathcal{M}$, this is a collection of 32 real scalar fields. The name "clock field" records their role as order-parameter coordinates; it does not by itself make four of them spacetime coordinates or produce a Lorentzian metric.

5 Soldering and Spacetime

The algebraic core gives a compact target space, not a spacetime. A four-dimensional Lorentzian spacetime must therefore be added as extra structure.

5.1 Minkowski blocks in the Albert algebra

The 2×2 Hermitian octonionic block

$$H_2(\mathbb{O}) = \left\{ \begin{pmatrix} a & x \\ \bar{x} & b \end{pmatrix} : a, b \in \mathbb{R},\ x \in \mathbb{O} \right\}$$

has determinant

$$\det \begin{pmatrix} a & x \\ \bar{x} & b \end{pmatrix} = ab - x\bar{x}.$$

Writing $a = X^0 + X^9$, $b = X^0 - X^9$, and $x = (X^1, \ldots, X^8)$, one obtains

$$\det = (X^0)^2 - (X^9)^2 - \sum_{i=1}^{8} (X^i)^2.$$

Thus $-\det$ has signature $(1, 9)$. The subalgebra $\mathbb{C} \subset \mathbb{O}$ gives $H_2(\mathbb{C}) \cong \mathbb{R}^{1,3}$.

Postulate 5.1 (Soldering)**.** There exists a rank-four soldering field

$$e : TX \longrightarrow H_2(\mathbb{C}) \subset H_2(\mathbb{O})$$

such that the spacetime metric is

$$g_{\mu\nu} = e^a{}_\mu e^b{}_\nu \eta_{ab}, \qquad \eta = \mathrm{diag}(-1, +1, +1, +1).$$

The choice of $d = 4$, the choice of $H_2(\mathbb{C}) \subset H_2(\mathbb{O})$, and the nondegeneracy of e are postulates.

Remark 5.2. This replaces any attempted derivation of $(1, 3)$ signature from the Killing form. The source of the indefinite form is the determinant on a chosen Albert-algebra block; the choice to use a four-dimensional complex subblock is additional physical input.

6 Gauge and Matter Sectors

6.1 Composite H-connection

For a local coset representative $g(x) \in E_6$, the Maurer-Cartan form decomposes as

$$g^{-1}\mathrm{d}g = A + \theta, \qquad A \in \mathfrak{h}, \quad \theta \in \mathfrak{m}.$$

Under a local H-change of representative, A transforms as an H-connection. Thus the coset geometry naturally supplies a composite $\mathrm{Spin}(10) \cdot \mathrm{U}(1)$ connection.

This observation is kinematical. A dynamical Yang-Mills term

$$-\frac{1}{4g_H^2} \mathrm{tr}(F_{\mu\nu} F^{\mu\nu})$$

must be included in the effective action or induced by integrating out specified massive fields. The coefficient is not fixed by coset geometry alone.

6.2 Matter embedding

Postulate 6.1 (Chiral matter)**.** Each family contains one chiral $\mathbf{16}$ of Spin(10), with the standard decomposition

$$\mathbf{16} \longrightarrow \mathbf{10}_{+1} \oplus \overline{\mathbf{5}}_{-3} \oplus \mathbf{1}_{+5}$$

under $\mathrm{SU}(5) \times \mathrm{U}(1)_X$. This contains one Standard Model family plus a right-handed neutrino.

Postulate 6.2 (Families and Yukawa sector)**.** Three generations are imposed by a family symmetry, for example $SU(3)_F$, and fermion masses arise from a conventional Spin(10) Higgs/Yukawa sector such as 10_H, 45_H or 54_H, and 126_H. These fields are part of the model data; they are not derived from the coset.

6.3 Anomaly bookkeeping

The complete $\mathbf{27}$ of E_6 is anomaly-free with respect to the standard E_6 embedding. In $H = \mathrm{Spin}(10) \cdot \mathrm{U}(1)$ notation,

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}.$$

Using $T(\mathbf{10}) = 1$ and $T(\mathbf{16}) = 2$ for Spin(10), the mixed U(1)-Spin(10)^2 anomaly of the full $\mathbf{27}$ is

$$(+1)T(\mathbf{16}) + (-2)T(\mathbf{10}) + (+4)T(\mathbf{1}) = 2 - 2 + 0 = 0.$$

The cubic $\mathrm{U}(1)^3$ anomaly is

$$16(1)^3 + 10(-2)^3 + 1(4)^3 = 16 - 80 + 64 = 0.$$

Remark 6.3. If the low-energy spectrum keeps only the $\mathbf{16}_{+1}$, the extra U(1) is no longer anomaly-free by itself. It must be broken, completed by the remaining E_6 states, or cancelled by an explicit mechanism.

6.4 Standard Model descent

The Standard Model does not follow merely from the existence of the $\mathrm{Spin}(10)\cdot\mathrm{U}(1)$ stabilizer. The descent is the usual Spin(10) grand-unified construction, included here as part of Theorem 2.3:

$$\begin{aligned}
E_6 \rightsquigarrow H, \qquad H &= \mathrm{Spin}(10) \cdot \mathrm{U}(1)_\psi, \\
H &\xrightarrow{\langle 45_H \rangle \text{ or } \langle 54_H \rangle} \mathrm{SU}(5) \times \mathrm{U}(1)_X \times \mathrm{U}(1)_\psi, \\
\mathrm{SU}(5) \times \mathrm{U}(1)_X \times \mathrm{U}(1)_\psi &\xrightarrow{\langle 126_H \rangle} \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y, \\
\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y &\xrightarrow{\langle 10_H \rangle} \mathrm{SU}(3)_c \times \mathrm{U}(1)_{\mathrm{em}}.
\end{aligned}$$

Here the first arrow is the vacuum selection already described by the coset, while the later arrows are Higgs-sector assumptions. Hypercharge is embedded in the standard $\mathrm{SU}(5) \subset \mathrm{Spin}(10)$ way, and the abelian factors orthogonal to Y must be lifted by the same Higgs sector, by Stueckelberg/Green-Schwarz data, or by other explicitly stated dynamics.

For one family,

$$\mathbf{16} \longrightarrow \mathbf{10}_{+1} \oplus \overline{\mathbf{5}}_{-3} \oplus \mathbf{1}_{+5}, \qquad \mathbf{10}_{+1} = (q, u^c, e^c), \quad \overline{\mathbf{5}}_{-3} = (d^c, \ell), \quad \mathbf{1}_{+5} = \nu^c.$$

Thus a chiral **16** contains exactly one Standard Model family plus a right-handed neutrino. A 126_H vacuum expectation value can give a large Majorana mass M_R to ν^c, and the usual type-I seesaw estimate is

$$m_\nu \simeq -m_D^T M_R^{-1} m_D.$$

Yukawa textures, proton-decay suppression, threshold corrections, and family structure remain independent phenomenological constraints on the postulated GUT sector.

7 Effective Dynamics

7.1 Sigma-model action

After Theorem 5.1, the clock fields propagate on the Lorentzian manifold (X, g). The leading two-derivative action for the coset field is

$$S_\sigma = \frac{f^2}{2} \int_X \mathrm{d}^4x \sqrt{-g}\, g^{\mu\nu} G_{AB}(\phi) D_\mu \phi^A D_\nu \phi^B,$$

where G is the positive E_6-invariant metric on EIII. The covariant derivative contains the H-connection. Couplings to the matter and Higgs sectors are additional effective-field-theory data.

At the level used in this paper, the low-energy action has the schematic form

$$S_{\mathrm{EFT}} = S_\sigma + S_{\mathrm{YM}} + S_{\mathrm{Higgs}} + S_{\mathrm{Yukawa}} + S_{\mathrm{grav}} + \cdots,$$

where the first term is fixed by the coset geometry up to its normalization, the middle terms are the postulated Spin(10)-to-Standard-Model sector, and the gravitational term is treated as an induced effective interaction.

7.2 Mass spectrum

At the algebraic vacuum, the 32 coset directions are Goldstone directions. Once the soldering sector, gauge fixing, and radiative corrections are included, the physical spectrum depends on the detailed effective action.

Assumption 7.1 (Working spectrum)**.** For estimates below, assume that four directions are tied to the soldered vierbein/gauge sector and that the remaining 28 real scalar modes acquire a common mass m.

Remark 7.2. This 28+4 split is not an H-invariant decomposition of the tangent representation $\mathfrak{m}$. It is a spacetime/soldering assumption. Therefore it should not be used as raw Spin(10) representation data in gauge beta-functions without specifying how the soldering sector breaks or gauges the relevant degrees of freedom.

7.3 Sakharov induced gravity

For n_s minimally coupled real scalars of common mass m, the one-loop heat-kernel contribution to the coefficient of R has the schematic form

$$\Delta M_{\mathrm{Pl}}^2 = \frac{n_s}{96\pi^2} m^2 \log \frac{\Lambda^2}{m^2} + \text{scheme-dependent local counterterms.}$$

With the working spectrum $n_s = 28$, this gives the estimate

$$\Delta M_{\rm Pl}^2 = \frac{28}{96\pi^2} m^2 \log\frac{\Lambda^2}{m^2} + \cdots .$$

This is a standard Sakharov estimate, not a UV-complete derivation of gravity. The cosmological constant term receives quartic divergences and must be tuned or addressed by additional dynamics.

7.4 Gauge running

For a gauge group G, the one-loop coefficient in the convention

$$\beta_g = -\frac{g^3}{16\pi^2} b_0 + \cdots$$

is

$$b_0 = \frac{11}{3} C_2(G) - \frac{2}{3} \sum_{\text{Weyl fermions}} T(R_f) - \frac{1}{6} \sum_{\text{real scalars}} T(R_s).$$

For $G = \mathrm{Spin}(10)$, $C_2(G) = 8$, $T(\mathbf{10}) = 1$, and $T(\mathbf{16}) = 2$. Three chiral **16** families contribute

$$\sum_f T(R_f) = 3\, T(\mathbf{16}) = 6.$$

The scalar contribution cannot be specified by the number 28 alone; it requires the actual Spin(10) representations of the light scalar fields. Consequently this paper does not quote a universal numerical b_0 for the full UCFT effective theory.

Conjecture 7.3 (Asymptotic completion)**.** *There may exist a UV completion in which the gauge-Yukawa sector is asymptotically free and the gravitational sector approaches an interacting fixed point. Establishing this requires a controlled truncation, regulator checks, and the complete field content. It is not proved here.*

8 Strings and Heterotic Correspondence

The compact space EIII is simply connected:

$$\pi_1(\mathrm{EIII}) = 0.$$

Therefore vortex strings do not follow from $\pi_1(\mathrm{EIII}) = \mathbb{Z}$. Any string sector must arise from additional gauging, a quotient by a broken U(1), or an independent topological sector.

Conjecture 8.1 (String sector)**.** *With suitable extra $U(1)$-breaking data, the effective theory may support finite-tension vortex solutions whose worldsheet degrees of freedom resemble a heterotic current algebra. This is a conjectural correspondence, not a duality proved by the coset construction.*

9 Summary

The cleaned UCFT proposal is the following.

1. The algebraic core is based on $J_3(\mathbb{O})_{\mathbb{C}}$ and compact E_6.

2. A stated E_6-invariant potential selects the projective rank-one orbit EIII.

3. The coset metric is Riemannian. Lorentzian spacetime is added by a soldering postulate.

4. Spin(10) matter, three generations, and the Higgs/Yukawa sector are model data, not consequences of the vacuum theorem.

5. Induced gravity and RG statements are effective estimates or conjectures until a complete field content and controlled truncation are supplied.

6. String/heterotic claims are conjectural and require extra topological structure beyond EIII.

This is now a mathematical-physics framework with clearly separated theorem, postulate, and conjecture layers. The next step is not to add more claims, but to prove or calculate one layer at a time: first the vacuum theorem and coset geometry, then the soldering sector, then the complete effective action and renormalization group.

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